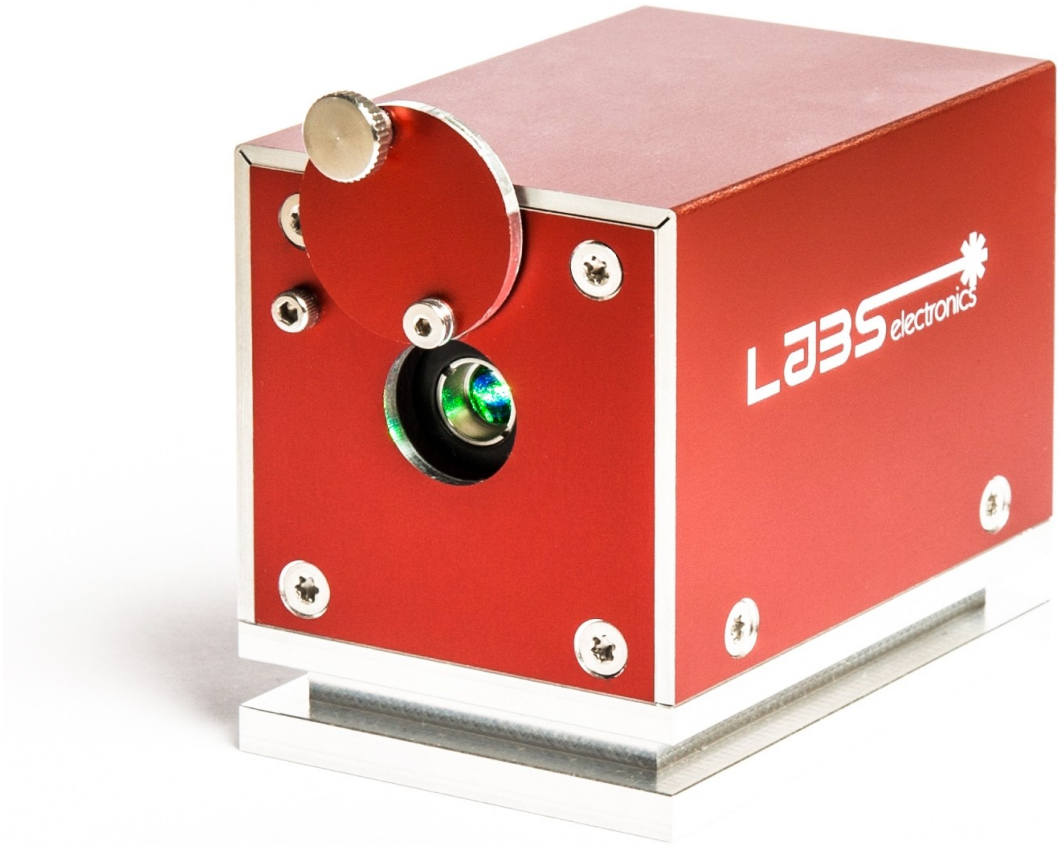
LABS electronics



DLnsec

Diode Laser with Sub-Nanosecond Digital Modulation

User Manual and Programming Reference

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Warning!

This device is not intended for private use. It reflects a scientific instrument for laboratory research purposes and may only be operated by qualified and trained scientific personnel .

Use by unqualified persons must be prevented.

Read this manual carefully and thoroughly before use and keep a printed version with the device.

VOID WARRANTY WARNING:

Some conditions lead to a loss of warranty. These conditions include but are not restricted to the following issues:

Opened device / broken seals:

DO NOT OPEN THE DEVICE! There are no user-maintainable parts inside. The warranty becomes void if screws are loosened, if the device has been opened or if any safety seal is broken.

Improper use:

HANDLE WITH CARE! Damage to the laser system due to a lack of proper care may lead to the loss of warranty claims on this product. Strictly mind voltage levels and polarity of power supply and trigger signals; don’t mix up input and output connections.

IMPORTANT NOTE FOR FIBER COUPLED MODELS:

Pictures shown in this manual present the standard free emitting version of the laser. Fiber coupled models do not have a collimating lens and do not carry a sticker marking the emitting aperture. The laser light is emitted from the end of the fiber. Before turning on the laser remove the end-cap from the fiber and attach it to a matching fiber coupler that is mounted rigidly on an optical table. Make sure that the beam path is free and safe.

HANDLE FIBER COUPLED MODELS WITH EXTRA CARE! The fiber **must not** be excessively stretched, bent, heated, or twisted.

1 Preface

Congratulations! With the DLnsec you have purchased a high quality scientific pulsed diode laser. The very small dimensions and the integrated controls make it suitable for remote-controlled setups and stand alone use even under the most difficult conditions and confined spaces.

The DLnsec is a sophisticated electronic high-end device with high reliability and a long durability. All components have been carefully checked and left our company in perfect condition. However, if anything is not to your complete satisfaction or if you have technical questions, please do not hesitate to contact us for support.

2 Safety Instructions and Warnings

2.1 Explanation

The DLnsec is a scientific device for laboratory use. During operation high power laser light is emitted. In order to avoid damage to health and equipment, please read these instructions carefully and keep them together with the device.
In the following chapters danger and warning notes are used to highlight specific topics that have to be considered to avoid any harm or personal damage.



Is used to indicate issues which may cause harm to persons or creatures directly or indirectly if the indication is violated.



Is used to indicate issues which may cause direct or indirect damage to the device or other laboratory equipment if the indication is violated.

2.2 Laser Safety



FOLLOW SAFETY INSTRUCTIONS - Before operating the DLnsec Laser System read this manual thoroughly and carefully. The safety instructions must be followed at all times. Always keep a printed version of this manual together with the system.



EMISSION OF LASER RADIATION – This is a Class 3B laser system emitting light in the visible (405nm – ca. 700nm) or invisible near infrared (> 700nm) range, depending on the model. The emitted wavelength and power are indicated in the spec sheet PDF file on the USB stick included with the laser and on the warning label attached to the top of the laser.



DANGER OF RETINAL DAMAGE – The radiation emitted by this laser system may cause serious damage to your eyes (danger of retinal damage) and skin by direct or reflected light. Always measure the emitted laser power with a calibrated power meter in cw mode before dealing with the laser. Don't trust your visual impression - especially light at the low (deep blue) and far (deep red) visible spectral range may appear less bright than it actually is.
Always take precautions to prevent exposure of eyes and skin to the beam. Always wear laser safety goggles suited for the wavelength and intensity of your particular laser system!



EMISSION OF SHORT LASER PULSES – This Class 3B laser system can emit cw laser radiation as well as short pulses with pulse widths on the order of nanoseconds and repetition rates above 10 MHz. A power meter will not be able to measure neither average nor peak power correctly under these conditions. Make sure the laser power is measured only in cw mode.
Although the peak power in a light pulse is not higher than the cw power, this is a possible source of danger because also the eye will see only the average power and thus underestimate the peak optical power in each pulse.



DO NOT USE AS LASER POINTER – This is a class 3B laser product for scientific use emitting high laser powers. Although it may look like a laser pointer this device must never be used for any other than laboratory purpose. It must be fixed to an optical table before use and be used in a protected environment and be made accessible to authorized and trained personnel only.



USE CERTIFIED PROTECTIVE EYE-WEAR – Always perform adjustment and alignment works with turned off and secured laser unit or at very low laser powers. Take special care of other persons that are not involved in the alignment works. Make sure everybody in the lab uses suited safety eye-wear. Keep in mind that others might not be aware of laser radiation being present.



USE WARNING MARKERS - High-performance lasers need imperatively great care and statutory warning markers have to be used with the device.



USE THE MECHANICAL SHUTTER – The DLnsec is equipped with a mechanical shutter to make sure no laser radiation is emitted unintentionally. Always keep the mechanical shutter closed when not operating the laser to prevent unintended laser emission. Also keep the mechanical shutter closed when branching the system to electricity.

2.3 General Safety Information



CHECK PARAMETER SETTINGS - To prevent damage to your Dlnsec laser system always ensure that the laser settings are set according to specification. In case of uncertainty reset laser parameters before putting the laser in operation state



USE ORIGINAL POWER SUPPLY ONLY - For safe operation please use the supplied mains cable and power supply only.



AVOID HAZARDOUS ENVIRONMENT - Do not operate the device in hazardous environment.



AVOID EXPLOSIVE ENVIRONMENT - Do not operate the device in explosive environment or together with easily flammable material.



DO NOT OPEN THE DEVICE - The user must not open the DLnsec Laser system. There are no user-maintainable parts inside. Opening may result in health-threatening danger and voids any warranty claims and we do not accept liability for any damages.



OPERATE AT LOW LASER POWER – As the used laser diodes have a limited lifetime you should not run the laser in a permanent emitting state or continuously on high power. To ensure performance and durability of the laser diode, use high laser powers only when necessary and for short time intervals. Reduce the power or switch of the laser whenever possible.

WARNING **AVOID ELECTROSTATIC DISCHARGE** - Protect the laser diode from short circuits, high currents and high voltages. As the laser diode is highly sensitive to voltage peaks and high currents, it may be destroyed by electrostatic discharge. Always ensure proper grounding of the device and people operating it.

WARNING **AVOID OPTICAL BACK COUPLING**- Always avoid optical feedback into the laser diode. Any reflection of the emitted laser radiation coupling back into the emitter can damage or destroy the laser diode.

WARNING **AVOID ELECTROMAGNETIC ENVIRONMENTS** - The device is not designed to be used in high electromagnetic radiation surroundings.

2.4 Laser safety glasses

IMPORTANT: Everybody involved in operating the laser must wear suitable safety glasses to protect their eyes from damage. It is important that the safety goggles have a **high enough optical density** for light with the **wavelength of your laser.** Note that laser safety glasses are usually strongly wavelength dependent and therefore it is important that the design wavelength of the glasses you intend to use match the wavelength of the laser. Do not use safety glasses whose design wavelength is not specified, that are designed for other wavelengths or whose optical density at your laser's wavelength is too small. Quality safety glasses are certified according to **DIN EN 207/208**. For our visible / NIR range lasers (**400 nm – 1500 nm**) with powers **up to 10 mW**, safety goggles need to have **OD4 or higher** to protect your eyes for up to 10 s of direct irradiation. For laser with **more than 10 mW** and **up to 100 mW**, the optical density has to be **OD5 or higher**. Ask your laser protection representative or protection eye-wear supplier for assistance when choosing suitable safety goggles for your laser.

3 Packaging Content

After unpacking your DLnsec, please check if all necessary components are contained and without damage

The DLnsec package contains:

- DLnsec Nanosecond pulsed diode laser
- Switching Power Supply
  Input: 100V-240V-60/50Hz 0.8A
  Output: 9V/3.0A
- USB Cable / IEEE 1394 with type A and B-Mini connectors.
- USB memory Device containing PDF Handbook and technical data
- External key switch
- Cable connecting key switch and laser

4 Quick Startup Procedure

For a very quick start with basic settings you can put the laser into operation with the following steps. Always make sure that you have read and understood the safety instructions first!

- Mount the device to an optical table using clamps or to a suited mounting plate using the threaded holes (metric M4/M6) on the bottom of the laser.

- Make sure the mechanical shutter is closed before connecting the laser to electricity.
- Connect the DLnsec to the external key switch using the included cable with two 2.5mm barrel connectors. **Do not try to connect the 2.1 mm barrel connector of the power supply directly to the 2.5 mm socket of the laser.** They do not fit!
- Plug the DC barrel connector into the correct 2.5mm plug of the key switch.
- Make sure the key switch is in OFF position.
- Connect the 3.5 mm jack to your interlock system. **The laser will only work when the interlock switch is closed, i. e. when there is a low-impedance connection between the the pins** of the 3.5 mm jack.
- Connect the power supply to mains.
- Make sure that the beam path is clear and safe operation is guaranteed.
- Turn key lock switch to ON position.
- Establish a USB connection to a serial port capable engine with a terminal emulator software (e.g. laptop computer using PuTTY; serial port parameters and PuTTY settings see sections 8.2 and 8.3).
- Test connection by sending `ERR?` or `*IDN`
- Check laser alignment and clear light-path.
- Set pulse power to approx. <n> % of maximum power sending `PWR <n>` (we strongly recommend starting with low power e.g. `PWR5`).
- Open mechanical shutter.
- Activate laser using command *`ON`
- Choose the desired mode of operation:
  - If you wish to use internal trigger send `INT`
  - For external trigger connect and activate trigger source and send `EXT`
  - For continuous wave operation send `LASE`
- To stop laser operation send `STOP`
- To turn the laser off send *`OFF`
- Close mechanical shutter.

5 Connections

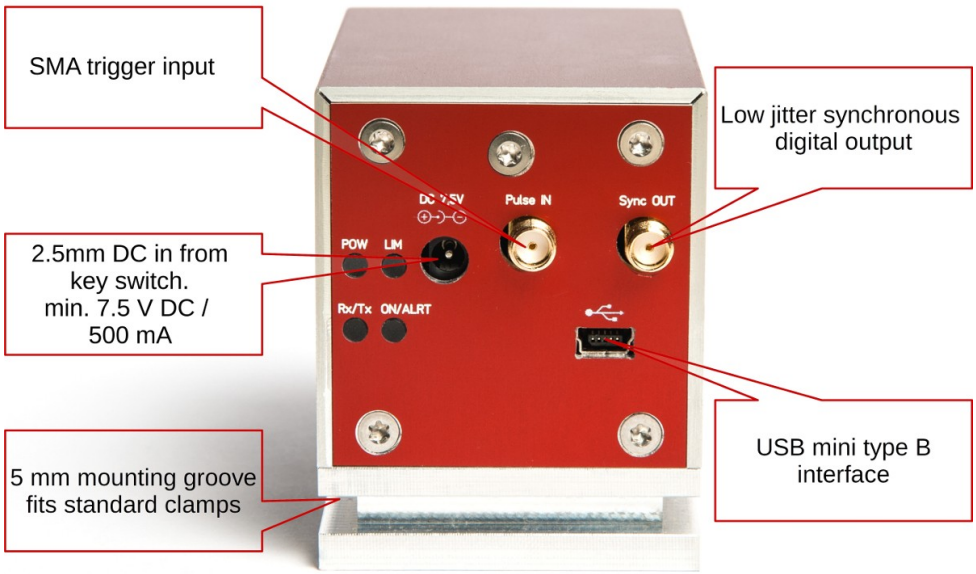


Figure 1: Back view of the laser system.

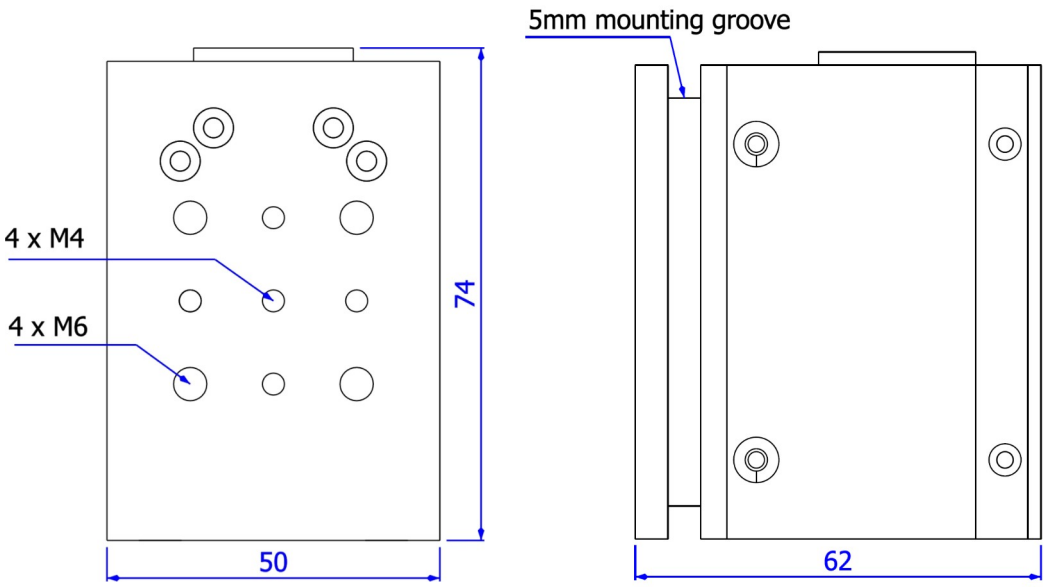
Status LEDs:

POW	[green or amber]	Indicates that the output stage is powered. (the color of this LED is model dependent)
LIM	[red]	Indicates that the system is being driven at its power limit. The laser diode is protected by a second regulator that sets a hard limit to the diode current. This regulator cuts the diode current to a save maximum level to prevent damage of the laser diode independent from the user settings. The laser has been calibrated to stay slightly below this limit even for nominally 100% output power. However, due to temperature change or other ambient conditions, such a situation can nevertheless occur. This condition is rather seldom and the warning simply indicates that the protection power level has been reached. This is in no way critical, however, the laser might perform better in terms of pulse shape and response if you reduce the current level by a few percent until the LIM LED goes off.
Rx/Tx	[green/orange]	Indicates that the system is receiving/sending data via the serial connection. Also blinks quickly 5 times when the system is powered or rebooted.
ON/ALRT	[green/red]	Indicates that the microcontroller unit is powered [green] or that a programming error has occurred [green+red at the same time] (the LED indicator looks yellowish). Usually this is due to miss-spelled commands or arguments sent to the device or the connection being operated at the wrong BAUDRATE. Place an `ERR?` command (chapter 10.2.1) to query the error message and empty the error memory. If `ERR?` returns a "`0; No error`" command without the ALRT LED being cleared, there might be a hardware error. In this case please contact us for support.

6 Installation

The DLnsec is supplied with multiple mounting possibilities.

It can be fixed to laboratory tables via M4 and M6 screws on the back side as well as by using standard clamps together with the mounting groove surrounding the base.

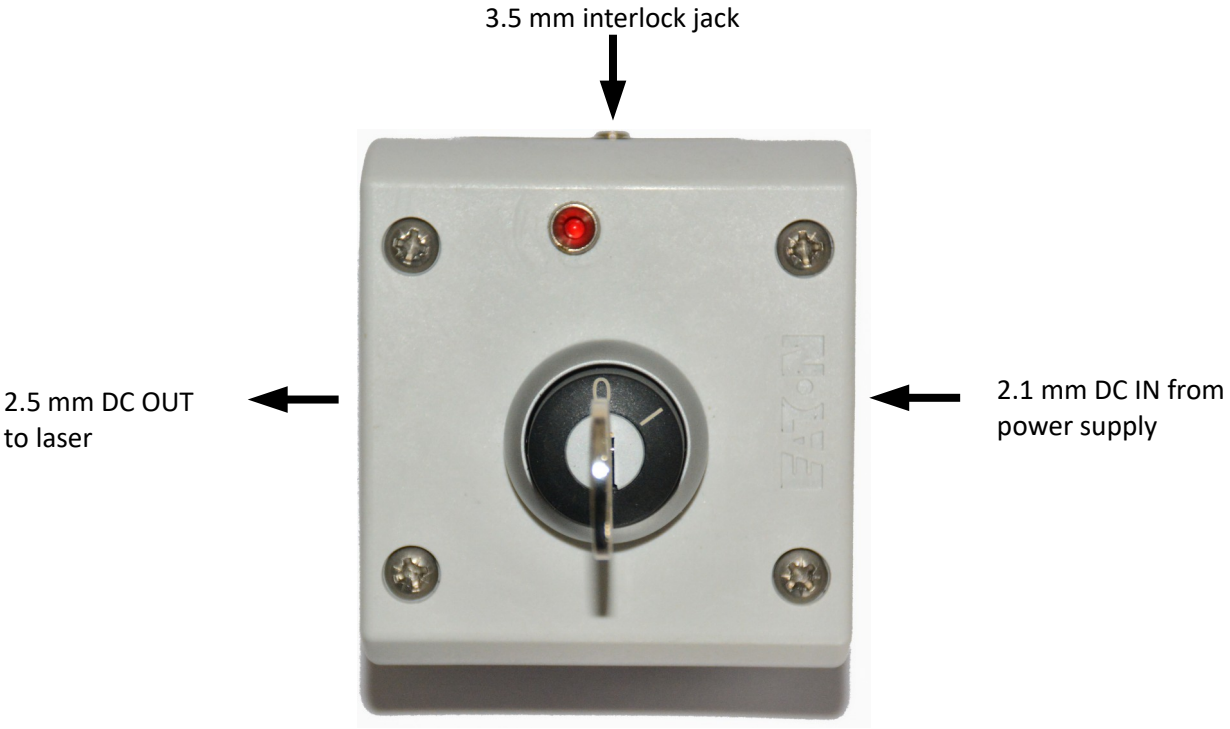


WARNING **DO NOT LOOSEN HEAD SCREWS ON THE BOTTOM** – Loosening the four countersunk head screws on the bottom side of the laser will irrevocably damage your laser system!

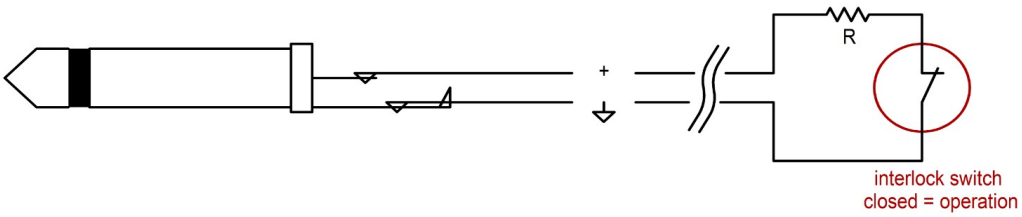
7 External Key-Switch & Interlock

The key switch has to be connected to the system as shown in the following figure and described in the text below.

DANGER **WARNING** **Do not try to override the key switch.** Any kind manipulation of any part of the key switch, the laser, the cables, or the connectors connecting them which is or could be able to render any of the protection measures ineffective must be omitted. Any kind of such manipulations voids any warranty claims and we do not accept liability for any damage to property, damage to persons or other living creatures.



- Connect the DLnsec to the external key switch using the included cable with two 2.5 mm barrel connectors. **Do not try to connect the 2.1 mm barrel jack of the power supply directly to the 2.5 mm socket of the laser.** They do not fit!
- Plug the DC barrel connector into the correct 2.5 mm plug of the key switch.
- Make sure the key switch is in “0” (=OFF) position.
- Solder the included 3.5 mm jack to your interlock system (e.g. door switch or similar) and plug it into the 3.5 mm socket of the key switch. The laser will only work when the interlock switch is closed, with a low resistance connection or short between the contacts of the 3.5 mm jack (see drawing below).



- To guarantee reliable operation, the total resistance R of the closed interlock switch and its leads must not exceed 10k Ohms.
- Plug the 2.1 mm DC plug coming from the power supply into the DC IN socket on the right side of the key switch.
- Make sure that the beam path of the laser is clear and that laser operation will be safe.
- Connect the power supply to mains and activate your system by turning the key switch in “1” (=ON) position.
- The led indicates that the laser is enabled.

8 USB Interface

8.1 Establishing a USB Connection

This device has no front panel controls. Therefore, to operate the laser you need to establish a USB connection between a PC / *MAC* and the laser using the included USB-A to B-mini cable at least once for some basic programming. All functions are controlled by commands that can be sent via a serial connection using the miniUSB port on the back-panel of the device. Attaching your device to a PC / *MAC* with a miniUSB / USB-A cable should allow you to use the connection immediately. However, on some Computers installing the correct driver might be necessary when you establish such a connection for the first time. On *Windows* 7 or higher the port should be recognized automatically. The port is provided by a very common built-in *FTDI* UART to USB bridge whose drivers should be included in every modern operating system or automatically found and installed when the device is connected while the computer is hooked up to the internet.

In the very unlikely case that the device is not available and drivers are not installed automatically please contact us for support.

The exact name of the device can differ depending on what software is running on your computer. However on Windows machines, the driver will establish as a virtual COM<xx> port, where <xx> depends on how many other (virtual or physical) COM ports exist on your machine. If you're not sure which port corresponds to your laser, disconnect and re-connect the device while watching the device manager.

There are many options to program your laser and many different software tools are available for all kinds of operating systems:

- Use a terminal program included in most and available for all Linux distributions or a commercial or a free terminal emulator like *PuTTY (see next section)* or similar to connect to the device. Any such program that is able to send plain text commands to a serial port is well suited. However, only single commands can be send. Scripting is not possible.
- For this purpose, e.g. *Python* and its serial communication abilities are better suited.
- As well as using serial communication via a *VISA* connection using *LabView*.
- *Matlab* also supports serial communication
- If you don't need scripting you could also use the terminal emulator included in the *Arduino* IDE to easily establish a bi-directional connection that allows you to communicate with the device.

Independent of which software you use, the baudrate and other parameters of the connection must be set according to the following paragraph.

8.2 Serial Port Settings:

- baudrate: 9600
- data bits: 8
- stop bits: 1
- parity: NONE
- flow control: XON/XOFF

8.3 Using PuTTY for Simple Communication

Though not really convenient for permanent use in the lab, using the free terminal client PuTTY to send plain text commands to and receive the replies from your device for testing purposes is sometimes useful. PuTTY is a small terminal emulator that can be downloaded for free from http://www.putty.org/.

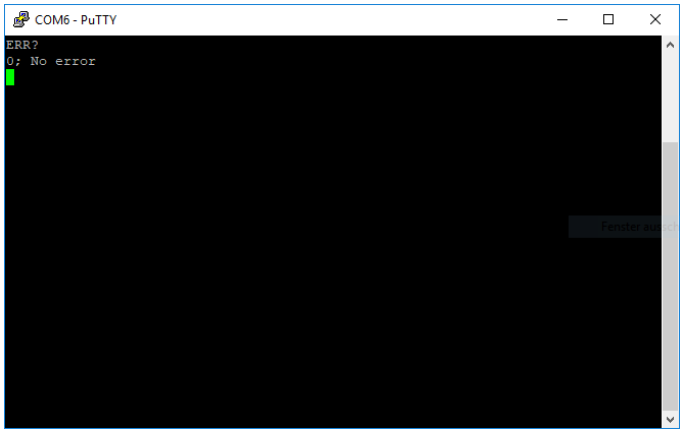


Figure 2: Answer received by PuTTY upon sending the ERR? command.

To establish a connection to the DLnsec the correct settings need to be made. Copy the settings documented in section 8.2 and save the settings before trying to establish the connection. All settings have to be correct. Otherwise the attempt to connect and send commands to the device will very likely fail even if the Rx / Tx LEDs stop blinking. If they do stop blinking, you should see a startup message as in 2. You can easily test communication by sending ERR? (See section 10.2.1).

8.4 Testing the USB Connection

If the settings are correct and you power the device while it is connected to the computer or you connect a computer while the device is powered, the connection should be almost instantaneously established. A safe way to make sure that the connection has been established is sending `ERR?` (see section 10.2.1). If a readable response is coming back from the device after a couple of milliseconds, the device is connected and can be controlled. Even receiving an error message other than `0,"No error"` means that communication settings are ok.

9 Internal Trigger

The DLnsec comes with an internal trigger generator that is programmable via the USB connection and can produce pulses lengths *t_pulse* down to 62.5 ns and repetition rates *f_rep* between 62 Hz and 63 kHz. Repetition rate and pulse width depend on two parameters: a width argument and a clock prescaler that determines the ratio between the basic auto-trigger clock and the system clock which runs at 16 MHz fixed frequency. Prescaler and width have a limited number of allowed values such that only certain combinations of *f_rep* and *t_pulse* are possible according to the rules discussed below. Also the accuracy and stability of the clock are both limited to about +/-0.5%. The drift with temperature is about +/-0.2% under normal working conditions. If more flexibility in the choice of pulse width and frequency and/or better stability are required, consider using a high quality external trigger source.

The clock prescaler can take five different values: 1, 8, 64, 256, 1024. These can be set using the `PRE <n>` command where <n> is one of the allowed values. Command arguments other than the allowed ones will issue a command argument error indicated by the ON/ALRT LED lighting orange/green. Depending on the prescaler setting the pulse repetition rate is f_rep = 16 MHz/(<pre> * 255).

The pulse width of the internal trigger generator can be set using the WID <n> command. Where <n> is the length of the pulse in units of prescaled clock cycles t_pulse = (<n>+1)*<m>/16 MHz. <m> and <n> are the prescaler / pulse width parameters, respectively. Allowed values for <n> are all integers in the range [0,255]. Depending on the output impedance of the external pulse generator, internal triggering may be possible or not when an external trigger source is connected. However to avoid interference and guarantee stable triggering, we recommend to disconnect any external trigger source when using then internal trigger. Continuous operation is usually not affected by connected external triggers but might be in rare cases. In such a case disconnect also the external trigger.

10 Programming Reference

In the following the commands that are understood by the laser are described.

10.1 General Remarks

- All commands are case sensitive.
- White spaces are ignored except for within keywords, i. e. `PWR50` is the same as `PWR 50` but `*OF F` is not the same as `*OFF`!
- Commands that relate to basic system functions have a mandatory leading star (`*`).
- Query commands are indicated by a trailing question-mark (`?`).
- To guarantee stable communication between the computer and the laser even in the presence of high bandwidth rf signals, we have chosen a rather low data communication rate. Keep in mind that data transfer takes a while.
- The device needs a couple of milliseconds (ca. 10ms) to parse incoming commands. If you use Python, LabView etc. for scripting, include a short delay between two consecutive commands.
- If you send a query command it takes a while for the device to answer. Give it a few milliseconds to reply[1].

10.2 List of Commands

10.2.1 General System Commands

ERR?

Reports the previous error. If no error has occurred, the device responds

```
0; No error
```

Otherwise an error message approximately following the SCPI 1999 standard is returned.

***OFF**

The output stage of the device is disabled. This mode allows to fully program the device but it does not emit light and synchronous trigger signals. This is the factory default mode at system startup.

1 If you are using python for scripting it might be necessary to wait up to 500ms before you can poll the reply with Python's <somevariable>=ser.read(<bytestoread>) command. We have sometimes encountered such behavior but suppose it is specific to Python because replies come as fast as expected when using other methods than Python.

*ON

Output state is activated. Upon this command the laser will immediately start operating with the specified parameters. If it is in external trigger mode (see below) it will start sending pulses depending on whether a trigger signal is present on the trigger input or not.

*RBT

Reboot the system and start with system defaults.

*IDN

Identify the system. It will send its model and serial number in the form `Dlnsecxxxx_yyyyy` where xxxx represents the central wavelength in nanometers and yyyyy is the serial number.

10.2.2 Save / Recall, User / Factory Settings

*SAV

Save user settings. This is useful if you want to start your device in a mode different from the factory preset. The command saves a number of parameters describing the mode of operation in a non volatile EEPROM memory which will be applied after a system power-on or reboot: the current setting, timer settings, current mode of operation (auto trigger/external trigger/continuous operation) and sleep or active mode). WARNING: The EEPROM has a limited number of 100.000 write cycles. This is pretty much and should be fully sufficient for a very long time, however when you are using scripting languages, LabView or similar to program the laser make sure you are not accidentally saving to the EEPROM in every run because that might degrade the memory very fast. You can query the number of write cycles that have already been performed by sending the `NSAV?` Command (see below).



It is the user's responsibility to avoid such extensive use of thee EEPROM memory. A laser system not working properly anymore due to a corrupted memory is not subject to any warranty and we do not accept any liability for any damages caused by a system with corrupted memory.

*RCL

Load and apply the previously saved user settings from the non-volatile memory.

*RST

Restore factory default settings. This overwrites all user settings by the initial factory defaults, applies them immediately to the system and saves them to the non-volatile memory, increasing the number of saves by one.

NSAV?

Query the number of EEPROM memory write cycles (max. 100.000 guaranteed, see above). If NSAV has reached 100.000 it may still be possible to save data to the memory but data integrity is not guaranteed anymore.



It is very likely possible to operate the laser even with a corrupted memory. Due to the distortion of the saved data, however, the system parameters after a system restart, a `*RBT`, or `*RCL` command might differ from the ones previously stored. Take special care that each system parameter is correct before triggering the laser in this case.



We do not guarantee for proper operation and accept no liability for any damages to the device, other things, persons or living creatures in the case of a corrupted memory.

10.2.3 Laser power settings

PWR <n>

Set pulse current to approx. <n> % of maximum power

PWR?

Get approx. Pulse power in % of maximum

10.2.4 Mode of Operation

EXT

Externally triggered mode (overrides LAS & INT)

LAS

CW laser operation (overrides EXT & INT)

INT

Internal trigger operation (overrides EXT & LAS)

STOP

Stop internal trigger, listen to external trigger (same as EXT)

10.2.5 Timer programming

PRE <P>

Set clock prescaler for internal trigger to <P> where <P>=[1,8,64,256,1024] such that f_{rep}=16MHz/256/<P>

PRE?

Get clock prescaler value <P> of internal auto trigger

WID <N>

Set pulse width of auto trigger to <N> where <N>=[0..255] such that t_{pulse}= 1/16MHz*<P>*<N+1>

WID?

Get pulse width <N>

10.3 Default State / User Preset

By factory default setting the DLnsec is programmed to a very low laser power and a non light-emitting state when branched to electricity. However it is possible to save user presets that allow to keep and recover the last state of the laser settings.



ADJUST PARAMETER SETTINGS – By using specific presets it is possible to put the laser immediately into a light-emitting state at high laser powers directly after powering the system. Always keep mechanical shutter closed when powering up the laser system. It is essential to check all system parameters before opening the mechanical shutter. In case of any doubt, reset the laser settings to default values before opening the mechanical shutter.

10.4 Check if the system is alive (and you mind asking ERR?)

HOWDY

Say Hello to your laser. Query the serial input to see its response.

11 Optical Adjustment

The optical output port contains an aspheric lens to collimate the beam. The lens position is factory set and usually no re-adjustment is necessary. However, for individual optical requirements the laser divergence can be adjusted by the user using a spanner wrench suited for M9 x 0.5 mounted optics (e.g. Thorlabs SPW301)

Operate the laser at low power, make sure that the beam path is clear and everybody involved is wearing suitable protective gear.

For adjustment, switch off the laser, insert the spanner wrench and adjust the position of the collimation lens by turning it.

Remove spanner wrench, turn on the laser and check the spot size on the screen. Repeat procedure until the desired spot size is achieved.

For optimal collimation the lens should be adjusted such that the smallest divergence is observed. Turning the lens from that point in either direction will increase the divergence of the beam in the far field.



DO NOT DRIVE THE LENS TOO FAR – Please take care not to drive the lens too far inwards as it might hit and damage the laser diode and its mount.

12 Specifications

12.1 Serial Number

Serial Number:	see attached spec.sheet

12.2 Electrical and Optical Properties

Light output

CW Power	see attached spec.sheet
center wavelength	see attached spec.sheet
beam diameter	see attached spec.sheet

Switching characteristics

Rise/fall times	see attached spec.sheet
Digital modulation bandwidth	>25 MHz

Digital inputs/outputs

TRIG input	SMA female, 50 Ohm terminated, 5V max.
TRIG high level	2.8 V
SYNC output	SMA female, 50 Ohm matched
SYNC output high level	>1.25 V on 50 Ohm
SYNC output rise/fall time	<2ns

General parameters

Laser diode current	1–300 mA (model dependent)
Power input	2.1 mm hollow jack
Power supply	7.5 V / 800 mA wall wart

Control interface

Connector	USB mini type B receptacle
Software driver	Only standard drivers required
Communication protocol	Virtual serial, plain text format

Dimensions

L x W x H	74 x 50 x 62 mm

12.3 Pulse characteristics

Measured with the standard 520 nm / 30 mW configuration. Results may differ for other types of laser diodes.

Pulse characteristics

Rise time	250 ps
Fall time	980 ps
shortest pulse (reaching full power)	300 ps FWHM

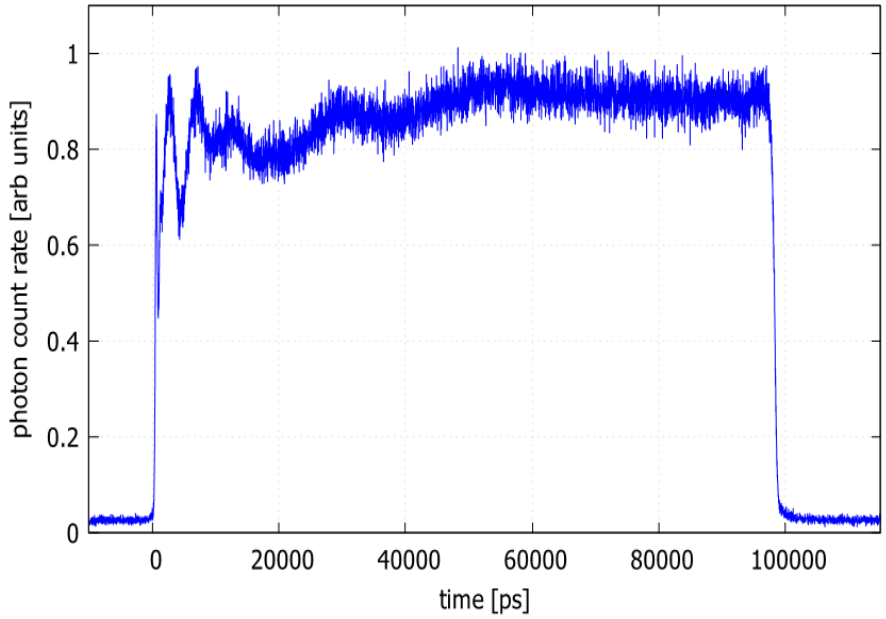


Figure 3: Typical pulse shape 520nm 30mW version.

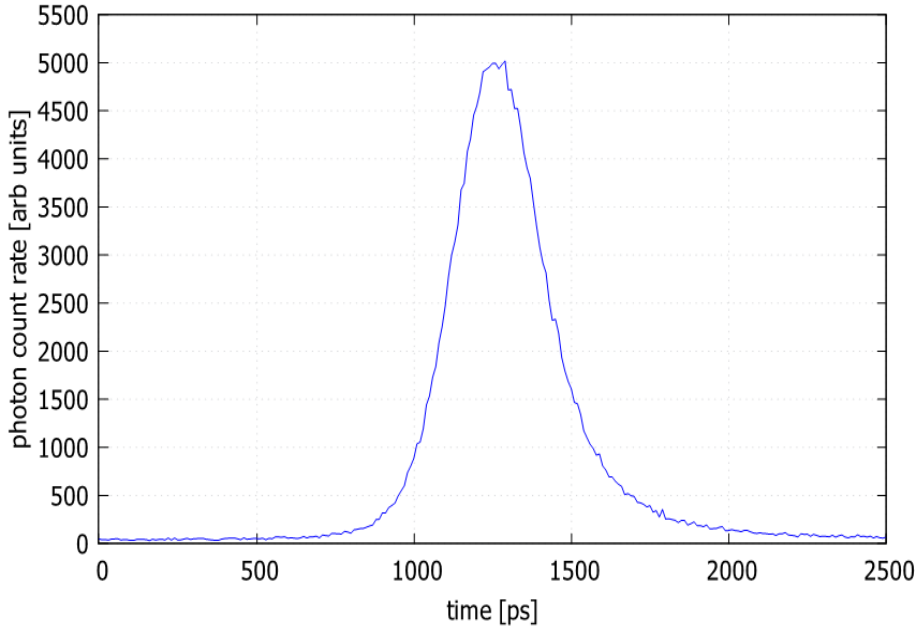
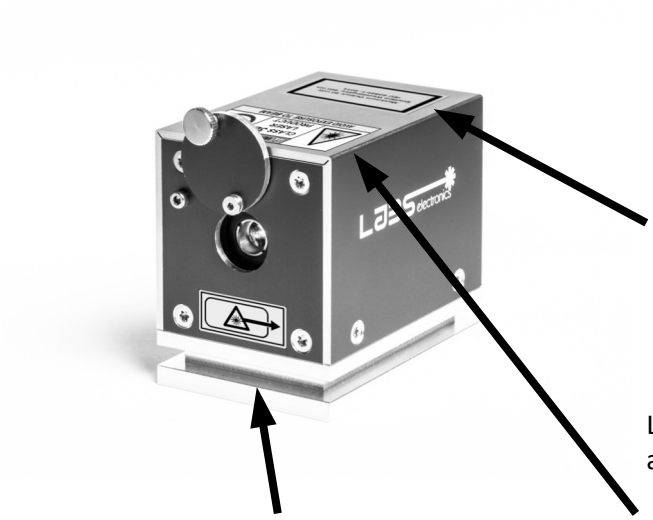
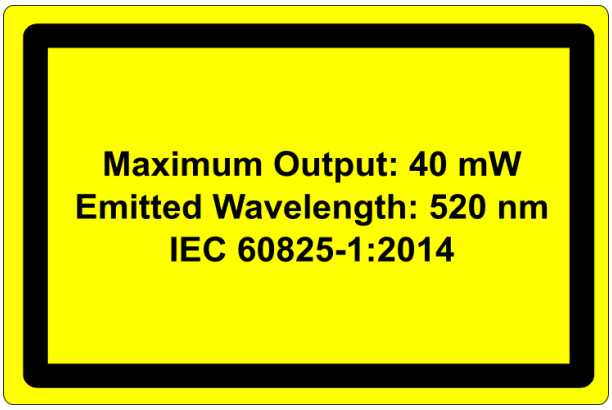


Figure 4: Shortest optical pulse reaching max. power. 300ps FWHM.

13 Warning Labels

The following warning labels are attached to the laser.





Label indicating the maximum laser output power and emission wavelength.



Label marking the light emitting aperture of the laser. The label is fixed below the aperture on the front side of the laser.



Label indicating the laser class and safety warnings. Alternatively, depending on the wavelength, this label also indicates the emission of invisible laser light:



14 Declaration of Conformity

EU Declaration of Conformity

in accordance with EN ISO 17050-1:2010

We, L.A.B.S. UG (haftungsbeschränkt) ,Lüderitzstraße 11, 71634 Ludwigsburg, Germany

in accordance with the following directive(s):

2014/30 EU	Electromagnetic Compatibility (EMC) Directive
2014/35/EU	Low Voltage Directive
2011/65/EU	Restriction of Use of Certain Hazardous Substances (RoHS)

herby declare that:

Model: DLnsec
Equipment: Nanosecond Pulsed Diode Laser

is in conformity with the applicable requirements of the following documents:

EN 61326-1:2013	Electrical Equipment for Measurement, Control and Laboratory Use – EMC Requirements
EN 61010-01:2010	Safety Requirements for Electrical Equipment for Measurement, Control and Laboratory Use
EN 62471-1:2009	Photobiological Safety of Lamps and Lamp Systems
EN 60825-1:2014-07	Safety of Laser Products

and which, issued under the sole responsibility of L.A.B.S. UG (haftungsbeschränkt), is in conformity with Directive 2011/65/EU of the European Parliament and of the Council of 8th June 2011 on the restriction of the use of certain hazardous substances in electrical an electronic equipment, for the reason stated below:

does not contain substances in excess of the maximum concentration values tolerated by weight in homogeneous materials as listed in Annex II of the Directive2014

I hereby declare that the equipment named has been designed to comply with the relevant sections of the above referenced specifications, and complies with all applicable essential requirements of the directives.

Ludwigsburg, 24 April 2018

Eric Schubert, CEO

Complies with 21 CFR 1040.10 and 1040.11
except for conformance with IEC 60825-1 Ed. 3.,
as described in Laser Notice No. 56, dated May 8, 2019

FCC

This device complies with part 15 of the FCC Rules. Operation is subject to the following two conditions: (1) This device may not cause harmful interference, and (2) this device must accept any interference received, including interference that may cause undesired operation.

Note:

This equipment has been tested and found to comply with the limits for a Class A digital device, pursuant to part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment. This equipment generates, uses, and can radiate radio frequency energy and, if not installed and used in accordance with the instruction manual, may cause harmful interference to radio communications. Operation of this equipment in a residential area is likely to cause harmful interference in which case the user will be required to correct the interference at his own expense.



Designed, developed and manufactured by

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Version / revision history of the DInsec manual

DLnsec_manual_V0.00	15.02.2018 first draft
DLnsec_manual_V1.00	05.03.2018 first version completed
DLnsec_manual_V1.01	27.06.2018 some typos corrected
DLnsec_manual_V1.02	03.07.2018 added CE conformity statement, FCC statement
DLnsec_manual_V1.03	09.09.2018 added powersupply; smaller typo corrections
DLnsec_manual_V1.04	24.04.2019 added key-switch chapter
DLnsec_manual_V1.05	19.11.2019 revised image and explanation of interlock function
DLnsec_manual_V1.06	13.12.2019 revised title page
DLnsec_manual_V1.07	24.03.2020 added "2.4 Laser safety glasses"; added 21 CFR 1040.10 and 1040.11 compliance notes; some minor typo corrections
DLnsec_manual_V1.08	08.09.2020 added remark about model dependence of color of POW indicator LED; added revision history of the manual
DLnsec_manual_V1.09	16.12.2020 minor layout correction
DLnsec_manual_V1.10	12.05.2021 added extra note for fiber coupled versions; UG replaced by GmbH; revised the preface